

Title: A Minimal AI Study Assistant for Research Support

Abstract:

This paper presents a lightweight AI study assistant designed for graduate students.

The system combines a large language model, simple agent task decomposition, and document retrieval.

Its goal is to help users understand papers, summarize academic content, and generate study plans.

Introduction:

Graduate students often need to read many research papers in a short time.

Traditional note-taking tools cannot answer questions based on uploaded documents.

A retrieval-augmented assistant can reduce reading time and improve research efficiency.

Method:

The system first extracts text from uploaded PDF files.

Then the text is split into small chunks and converted into vector embeddings.

A FAISS index stores these vectors and retrieves the most relevant chunks for each user question.

The retrieved context is passed to a language model to generate grounded answers.

A simple rule-based agent decides whether the task is question answering, summarization, or study plan generation.

Contributions:

First, the paper proposes a minimal architecture for an AI learning and research assistant.

Second, it integrates PDF parsing, vector retrieval, and language model answering in one Streamlit interface.

Third, it adds a simple agent that can decompose user tasks into summary, planning, and QA steps.

Fourth, the system is easy to extend with more advanced RAG and multi-agent capabilities.

Conclusion:

The prototype demonstrates that a small and clear system can support academic reading and learning.

Future work will improve retrieval quality, add citation tracing, and support multiple documents.